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AI in EFL blended learning: impact on speaking performance and learning resilience

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Abstract

This study investigates the impact of AI-assisted blended instruction on EFL learners' speaking performance and learning resilience. Grounded in Social Cognitive Theory, the Technology Acceptance Model, and Sociocultural Theory, the research addresses a growing need to understand how AI tools shape not only linguistic accuracy but also learners' psychological adaptability. Adopting a quasi-experimental design, the study compared an experimental group using ChatGPT and Gemini within a task-based learning framework to a control group receiving conventional instruction over an 18-week period. Quantitative analyses revealed significant gains in pronunciation and accuracy among the experimental group, with coherence showing marginal improvement and fluency remaining unchanged. In terms of resilience, learners demonstrated marked increases in metacognitive and social resilience, whereas ego resilience remained unaffected.

These findings suggest that while AI tools can effectively support linguistic development and foster strategic learning behaviors, they may be less effective in cultivating emotional adaptability. The differentiated outcomes highlight the importance of aligning specific AI functionalities with targeted pedagogical goals rather than assuming uniform benefits. This study contributes a nuanced perspective to the discourse on AI in language education by uncovering domain-specific effects and clarifying the boundaries of AI's pedagogical influence. It calls for future research to explore the long-term impacts of AI integration on learner autonomy, motivation, and affective growth, thereby paving the way for more intentional, theory-driven applications of AI in EFL contexts.

Keywords: Artificial intelligence, Blended learning, Lower-intermediate EFL learners, Speaking performance, Learning resilience

Introduction

English as a Foreign Language (EFL) education is undergoing a transformative shift due to the integration of advanced technologies, particularly artificial intelligence (AI), in language instruction. This evolution is especially apparent in efforts to improve speaking proficiency among lower intermediate learners, a group that frequently faces both linguistic limitations and psychological barriers. While traditional classroom instruction

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remains foundational, AI enhanced blended learning has emerged as a promising alternative that combines structured guidance with adaptive, learner centered support (Namaziandost & Rezai, 2024). AI powered educational tools now enable real time feedback, personalized dialogue practice, and individualized pronunciation training. These features are reshaping how learners engage with spoken language.

One crucial but often underexplored dimension of language learning is resilience, defined as the ability to persist, adapt, and progress despite setbacks (Wang, 2021). For lower intermediate EFL learners, resilience is particularly critical, as they must navigate not only grammatical complexity but also emotional and metacognitive demands (Qiao & Zhao, 2023). Conventional teaching methods, while offering structured environments, may not fully address the psychological aspects of language learning, such as learners' emotional engagement, perceived competence, sense of autonomy, and coping strategies, which are essential for developing resilience (Alahmari & Alrabai, 2024). This challenge is especially salient in speaking instruction, where affective variables play a decisive role in communicative development.

Although blended learning models have demonstrated benefits for vocabulary learning and reading comprehension (Chiu, 2021; Hashemi & Na, 2020; Ortega Ruipérez, 2022), relatively few studies have examined the specific impact of AI enhanced instruction on speaking subskills, namely pronunciation, fluency, coherence, and accuracy, all of which are essential for real world communication. Even fewer have investigated how AI supported environments may influence learners' resilience, including its metacognitive, emotional, and social facets.

This study addresses these research gaps by adopting a task-based design in which different AI tools were strategically integrated according to instructional purpose. Specifically, Google Gemini was used to support idea development and pre speaking planning, while a ChatGPT based chatbot served as the primary tool for real time oral practice and feedback. This pedagogical alignment enabled students to experience multiple AI interfaces while ensuring consistency in speaking task execution. All participants engaged with these tools under unified learning conditions, thereby minimizing device variability and enabling focused comparisons.

By simultaneously examining the impact of AI supported instruction on both speaking proficiency and learning resilience, this study contributes new empirical insights to the evolving discourse on technology mediated language learning. In doing so, it clarifies how different AI applications can be purposefully employed to enhance learner performance and persistence. Its findings aim to inform instructional design and curricular decision making, particularly in the implementation of AI tools that support both linguistic competence (pronunciation, fluency, accuracy, and coherence) and psychological readiness (confidence, anxiety management, and willingness to communicate) essential for successful EFL communication.

Literature review

This section presents a comprehensive review of literature related to AI-enhanced EFL learning, speaking performance, and learning resilience. The review synthesizes findings from previous research to establish the theoretical and empirical foundation for the present study.

Speaking performance in AI-enhanced EFL learning

Challenges at lower-intermediate level

Speaking performance presents significant challenges for lower-intermediate EFL learners, particularly in developing fluency and confidence in oral communication (Ebadijalal & Yousofi, 2023). Research consistently identifies several interconnected obstacles that impede progress at this proficiency level.

Vocabulary and fluency limitations Albelihi (2022) conducted a quasi-experimental study with 60 intermediate Saudi EFL learners (aged 12–17) using a pretest–posttest design to examine the impact of lexical chunks instruction on speaking fluency. The experimental group ($n = 30$) received 18 sessions of explicit lexical chunks training over the intervention period, while the control group ($n = 30$) followed traditional instruction using the Headway Book 3 coursebook with a grammar-translation approach. The experimental group showed significant within-group improvement in speaking fluency from pretest ($M = 9.981$, $SD = 1.4950$) to posttest ($M = 12.021$, $SD = 1.3752$), with a statistically significant difference ($p < .001$). However, when comparing between groups in the posttest, no significant difference was found ($p = .397$), as the control group also achieved improved scores ($M = 10.05$, $SD = 1.9686$) compared to the experimental group ($M = 12.021$, $SD = 1.3752$). This finding is particularly relevant to the present study as it demonstrates that even targeted vocabulary interventions may not significantly outperform traditional methods, suggesting the need for more comprehensive approaches that address multiple dimensions of speaking challenges simultaneously.

Psychological and cognitive processing challenges Utami & Amalia (2024) investigated 21 first-year university EFL students using a qualitative approach with Likert scale questionnaires and interviews to examine challenges in oral presentations. Their findings identified two main challenge categories: personal traits and presentation skills. Regarding personal traits, most students experienced anxiety, nervousness, and feelings of being unprepared during oral presentations. For presentation skills, students demonstrated particular difficulties with pronunciation, maintaining fluency, and organizing thoughts in real-time speech production. The connection between this research and Albelihi's work is significant: while Albelihi's lexical intervention showed limited effectiveness compared to traditional instruction, Utami and Amalia revealed that speaking difficulties extend beyond vocabulary to encompass psychological barriers and cognitive processing demands. This is precisely the multidimensional challenges that the present study aims to address through both linguistic skill development and learning resilience enhancement.

Task complexity and strategic breakdown Jouibar et al. (2021) examined 61 lower-intermediate Iranian EFL learners through controlled speaking tasks of varying complexity levels. Their analysis revealed that less than 10% of students' clauses included appropriate code-switching during speaking tasks, with performance deteriorating significantly as task complexity increased, evidenced by increased hesitation and reduced fluency. This finding directly connects to the theoretical framework of the present study: it demonstrates that lower-intermediate learners lack the strategic resources

and adaptive capacity to maintain performance under cognitive pressure—challenges that require both linguistic competence development and psychological resilience building.

These three studies collectively reveal that lower-intermediate speaking challenges operate at multiple interconnected levels: lexical instruction shows limited superiority over traditional methods (Albelihi, 2022), psychological barriers including anxiety and confidence issues persist across learners (Utami & Amalia, 2024), and strategic-cognitive breakdowns occur under increased task complexity (Jouibar et al., 2021). However, none of these studies examined how technological intervention might simultaneously address these interconnected challenges. The present study builds on this foundation by investigating whether AI-enhanced blended learning can provide the comprehensive support needed to address both the linguistic and psychological dimensions of speaking development at the lower-intermediate level.

AI applications in speaking instruction

Recent research has demonstrated varying and complex relationships between AI applications and English speaking instruction, revealing differential effects across anxiety, performance, and self-regulation domains among EFL learners.

AI impact on anxiety and performance: mixed findings El Shazly (2021) conducted an 8-week quasi-experimental pretest–posttest study with 48 undergraduate EFL participants in Egypt to examine the role of AI applications in speaking practice for foreign language anxiety (FLA) management. The study employed a 33-item FLA questionnaire to assess anxiety levels and used an interaction-enhanced public version of the IELTS speaking evaluation rubric for oral proficiency assessment through role-playing activities. Results confirmed that conversationally enhanced AI chatbots significantly improved linguistic output gains; however, learners' speech-related anxieties were not reduced following interactions with the chatbots. Surprisingly, the use of AI chatbots in interactive activities slightly intensified learners' FLA despite the performance improvements. This finding is crucial for the present study as it demonstrates that AI tools may enhance linguistic performance while simultaneously creating different anxiety dynamics, highlighting the need to examine both performance outcomes and psychological resilience concurrently.

Comprehensive impact on speaking and self-regulation Qiao and Zhao (2023) conducted a more extensive randomized controlled trial with 93 Chinese EFL students over 13 weeks, with participants randomly assigned to either an experimental group ($n = 47$, age: $M = 20.2$, $SD = 1.8$; 45.5% female) receiving AI-based instruction using the Duolingo application, or a control group ($n = 46$, age: $M = 20.1$, $SD = 1.7$; 37.5% female) receiving traditional instruction. Speaking skills were assessed using IELTS speaking tests evaluating fluency, vocabulary, accuracy, and pronunciation, while self-regulation was measured using the Self-Regulated Language Learning Questionnaire (SRLQ). Results demonstrated significant improvements in all four speaking components: fluency ($B = 0.65$, $p = .006$), vocabulary ($B = .57$, $p = .034$), accuracy ($B = .46$, $p = .013$), and

pronunciation ($B = .42, p = .025$), as well as enhanced self-regulation ($B = .51, p = .018$). This comprehensive study directly supports the present research's dual focus on both speaking performance and learning resilience, providing empirical evidence that AI tools can simultaneously develop linguistic competence and self-regulatory capabilities.

Social integration in AI-assisted learning Zou et al. (2023) investigated how social network-based interactions could enhance AI applications in supporting speaking practice through a quasi-experimental design with 70 students from different Chinese universities during a five-week summer vacation. Participants were divided into experimental and control groups, with the experimental group encouraged to engage in interactive activities when practicing with AI apps, while the control group used AI speaking apps without social interaction. Data were collected through questionnaires, semi-structured interviews, and pre- and post-tests.

Results demonstrated that AI chatbots significantly boosted speaking proficiency when integrated with social interactions, bridging individual AI use with collaborative learning contexts. Participants in the experimental group showed greater engagement and improved speaking outcomes compared to those using AI apps in isolation. This finding connects to the present study's framework by demonstrating how AI effectiveness can be enhanced through social integration, informing the design of AI-enhanced blended learning that maintains classroom social structures while incorporating technological support.

Practical implementation in vocational contexts Most recently, Ahyarudin & Jamilah (2024) conducted action research over two cycles with 35 hospitality management students (25 females, 10 males) in the 2023/2024 academic year, integrating AI-based Instructional Tasks (AI-IT) into the Genre-Based Approach (GBA). Using qualitative methods including interviews, observations, and document analysis following Miles and Huberman's framework, the study employed AI tools such as Deep English AI Chatbot, Murf AI Text-To-Speech (TTS), and Speech Notes AI Speech-To-Text (STT). Findings revealed that AI-IT significantly enhanced students' vocabulary, pronunciation, and confidence, particularly in transactional language relevant to their field, while also increasing classroom engagement and active participation. This study provides direct practical evidence for the feasibility of integrating AI tools within structured pedagogical frameworks, specifically informing the present study's task-based implementation approach within blended learning environments.

These four studies reveal a clear methodological progression in AI speaking research: from anxiety-focused investigations with small samples (El Shazly, 2021: $n = 48$) to comprehensive randomized trials examining multiple outcomes (Qiao & Zhao, 2023: $n = 93$), social integration studies (Zou et al., 2023), and practical implementation research in specific contexts (Ahyarudin & Jamilah, 2024: $n = 35$). However, no previous study has systematically examined how AI integration affects both specific speaking performance subcomponents (pronunciation, fluency, accuracy, coherence) and multidimensional learning resilience (ego, metacognitive, social) within the same comprehensive intervention. The present study builds on this foundation by addressing this methodological

gap through simultaneous assessment of linguistic and psychological outcomes in an 18-week semester-long intervention, providing a more holistic understanding of AI's educational impact.

Learning resilience and AI-enhanced blended teaching

Learning resilience in EFL context

Learning resilience, conceptualized as a dynamic process of positive adaptation despite significant adversity in language learning contexts, has emerged as a critical factor in EFL achievement (Kim & Kim, 2016). Drawing from resilience theory in psychology (Luthar et al., 2000) and adapted for language learning contexts, research indicates that resilience comprises multiple dimensions including ego resilience, cognitive resilience, and social resilience, which collectively enable learners to overcome challenges in foreign language acquisition (Guo & Li, 2022).

Conceptual foundations and multidimensional framework Wang et al. (2021) conducted a comprehensive conceptual review of positive psychology in second/foreign language learning and teaching, examining seven instances of positive psychology variables including resilience. Their analysis emphasized that resilience serves as both a protective factor against negative emotions and a facilitator of positive psychological resources in language learning. According to their framework, resilience encompasses both internal processes (individual characteristics) and external processes (environmental supports) that enable healthy adaptation during difficulties, trauma, threats, and daily distress. The review noted that resilience has been associated with different L2 educational outcomes such as learning motivation, well-being, and teaching enjoyment, despite being under-researched in the language education literature.

Empirical evidence of resilience-motivation relationships Kim & Kim (2021) conducted a structural equation modeling study investigating the relationship between L2 learning motivation and resilience and their impact on motivated behavior and L2 proficiency. Their research revealed that resilience explained 70.6% of the variance in learners' engagement levels, suggesting its crucial role in sustaining long-term language learning efforts. The study found that resilient learners demonstrated greater capacity to maintain engagement in language learning activities even when faced with challenges or setbacks. This finding is particularly significant for the present study as it provides empirical evidence that resilience directly influences learners' ability to sustain engagement—a key component that the current research aims to develop through AI-enhanced instruction.

Cross-cultural validation and measurement development Wang et al. (2022) conducted a multi-dimensional cross-cultural study developing resilience among Chinese and Iranian EFL teachers, employing both quantitative and qualitative methods. Their research confirmed that resilience is not merely a static trait but rather a dynamic characteristic that can be developed and enhanced through positive learning experiences. The study highlighted that socio-affective factors, such as peer relations, teacher support, and institutional expectations, played crucial roles in fostering resilience among EFL educators. This cross-cultural validation is directly relevant to the present study's context with Chi-

nese EFL learners, providing evidence that resilience development strategies can be effectively implemented across different cultural settings.

Academic resilience and motivation interconnections Zhang (2022) conducted an empirical study with 482 Chinese EFL learners examining the relationship between resilience and academic motivation using Spearman's rho correlation and multiple regression analyses. The study employed a 40-item resilience scale (Cronbach's $\alpha = .971$) and a 28-item motivation questionnaire (Cronbach's $\alpha = 0.883$). Results revealed a significant relationship between resilience and motivation ($r = -.153, p = .001$), with interjected regulation ($B = .20, p < .001$) and external regulation by attendance ($B = .15, p = .01$) proving to be the best predictors of learners' resilience. The study concluded that learners with high motivation to learn can exert more effort and overcome demotivating factors when they possess sufficient resilience. This empirical evidence directly supports the present study's theoretical framework by demonstrating the quantifiable relationship between resilience and specific motivational components in Chinese EFL contexts.

These studies collectively reveal that resilient EFL learners share several common characteristics: they demonstrate better adaptive strategies, maintain higher engagement levels, and show greater persistence in achieving their language learning goals despite encountering obstacles (Kim & Kim, 2021; Wang et al., 2021). Importantly, research has shown that academic resilience can be cultivated through appropriate pedagogical interventions and supportive learning environments, making it a valuable target for educational enhancement efforts in EFL contexts (Wang et al., 2022). However, no previous study has systematically examined how AI-enhanced blended learning might simultaneously foster both speaking performance development and multidimensional resilience building. The present study addresses this gap by investigating whether technological intervention can provide the comprehensive support needed to develop both linguistic competence and psychological resilience among lower-intermediate EFL learners.

The interconnection between learning resilience and speaking performance

While speaking performance and learning resilience are often investigated separately in EFL research, their development is deeply interwoven, especially for lower-intermediate learners who frequently encounter affective, cognitive, and social barriers during oral language tasks. Speaking, as one of the most anxiety-inducing language skills, requires learners to manage the fear of making mistakes, perceived social judgment, and the cognitive load of real-time language processing (Ebadijalal & Yousofi, 2023; El Shazly, 2021). These barriers often lead to avoidance behaviors or reduced participation, directly impacting speaking performance.

Learning resilience, defined as learners' capacity to adapt, persist, and recover from challenges in the language learning process (Wang, 2021), plays a critical role in navigating these barriers. Resilience enables learners to tolerate ambiguity during communication, manage the discomfort of errors, and continue participating in speaking tasks despite temporary setbacks (Kim & Kim, 2021). Alahmari & Alrabai (2024) empirically demonstrated that resilience is a significant predictor of classroom engagement, which is a prerequisite for developing communicative competence.

The connection between resilience and speaking performance can be further explained through Social Cognitive Theory (Bandura, 1986), where self-efficacy and self-regulation—key components of resilience—shape learners' willingness to engage in and persist with speaking tasks. Learners with higher resilience are more likely to initiate and sustain participation in oral activities, seek feedback actively, and use errors as learning opportunities, all of which contribute to incremental improvements in fluency, accuracy, and coherence (Qiao & Zhao, 2023). Resilient learners also tend to set clearer goals for their speaking development and monitor their progress, aligning with the metacognitive dimensions of resilience highlighted by Guo & Li (2022).

Conversely, engaging in structured speaking activities can foster resilience by creating opportunities for learners to experience manageable challenges, receive immediate corrective feedback, and develop adaptive strategies to handle communication breakdowns. Each successful interaction, even with minor errors, can build learners' confidence and perceived competence, essential affective components of resilience (Zhang, 2022). Through repeated cycles of speaking practice, feedback, and self-reflection, learners develop a growth-oriented mindset, gradually reframing speaking challenges as opportunities rather than threats.

This reciprocal relationship implies that speaking performance and learning resilience should be investigated together rather than in isolation. Improvements in resilience can enable learners to engage more fully and productively in speaking tasks, while gains in speaking competence can strengthen learners' confidence and persistence in language learning. For AI-enhanced blended learning contexts, understanding this interplay is particularly crucial. AI tools, when integrated with task-based speaking practices, can simultaneously address linguistic accuracy and foster learners' psychological readiness by providing structured challenges, adaptive feedback, and opportunities for self-directed learning. Investigating both constructs together enables a holistic understanding of how AI can support learners not only in developing communicative competence but also in building the resilience necessary to sustain long-term language learning engagement.

Impact of AI-enhanced blended teaching

Recent empirical research demonstrates that AI-enhanced blended teaching can positively impact student learning outcomes through various mechanisms, including personalized feedback, adaptive content delivery, and enhanced student engagement (Holstein et al., 2018; Matazu, 2024). Analysis of multiple studies reveals both the potential benefits and implementation challenges of integrating AI into educational settings, with emerging evidence suggesting that AI tools may contribute to learning resilience development.

Matazu (2024) conducted a quasi-experimental study with 70 undergraduate students comparing AI-blended learning, AI-personalized learning, and traditional classroom instruction. Results showed that the AI-blended learning group achieved significantly higher gains in academic performance, with specific improvements in post-test scores (experimental group: $M=12.26$, $SD=4.07$ vs. control group: $M=10.34$, $SD=4.64$) and attitudes toward learning. These findings suggest that AI tools may enhance learner

persistence and adaptive capacity by providing immediate feedback and personalized support that helps students overcome academic challenges.

Holstein et al. (2018) found positive effects in their study of 286 middle school students, where providing teachers with real-time AI analytics about student learning and behavior led to improved student outcomes, particularly benefiting students with lower prior achievement. This targeted support for struggling learners indicates that AI systems may build resilience among academically vulnerable students by identifying difficulties early and enabling timely interventions.

The importance of comprehensive teacher support and technological infrastructure for successful AI-enhanced blended learning implementation emerges clearly in recent findings. Sanders & Mukhari (2024) conducted qualitative interviews with South African higher education lecturers, finding that successful blended learning implementation depends heavily on institutional support, technological infrastructure, professional development, and adequate time allocation. Pan (2024) empirically validated an AI-enhanced teaching model that emphasizes the need for both technical and pedagogical preparation to ensure effective implementation. These findings highlight that resilience development requires not only individual learner support but also systemic institutional resilience through robust support structures.

The collective evidence indicates that AI-enhanced blended teaching can improve educational outcomes when properly implemented with adequate support structures. However, the specific mechanisms through which AI influences different dimensions of learning resilience—including metacognitive, social, and ego resilience—remain underexplored, highlighting the need for targeted research that explicitly examines these relationships.

Theoretical framework

This study integrates three complementary theories that explain how technology-enhanced instruction influences both cognitive and affective dimensions of language learning.

Social cognitive theory

Bandura (1986) explains how learners develop self-efficacy through observational learning and self-regulation processes. The theory posits that immediate feedback enables learners to observe correct models and self-correct, strengthening their belief in their abilities. Recent studies have applied this theory to AI-enhanced language learning contexts, demonstrating that AI-mediated feedback can enhance learner self-efficacy (Lai, 2025). This theory guided the selection of learning resilience as a key outcome variable, as resilience encompasses the self-regulatory processes crucial for sustained learning engagement.

The technology acceptance model

Davis (1989) predicts that users' perceptions of technology usefulness and ease of use determine their acceptance and continued engagement with technology. This framework has been successfully applied to educational technology contexts, including AI-enhanced language learning, where studies confirm that positive technology perceptions

lead to higher engagement and persistence (Schmidt & Strasser, 2022). This model guided the selection of ChatGPT and Gemini based on their established usability among university students.

Sociocultural theory

Vygotsky (1978) emphasizes that learning occurs through social interaction and cultural mediation, where tools and signs serve as mediational means to transform mental functioning. Contemporary researchers have extended this framework to examine how digital technologies can function as mediational artifacts in language learning contexts (Lantolf & Thorne, 2006). This theory informed the integration of AI tools within existing classroom social structures, ensuring technology supplemented rather than replaced peer interaction and teacher guidance.

These theories collectively provide a framework for understanding how technology-enhanced instruction may influence language learning outcomes. This theoretical synthesis guides the investigation of how AI-enhanced blended instruction affects both speaking performance components and learning resilience dimensions among lower-intermediate EFL learners.

Research gap

Despite growing interest in AI-enhanced language instruction, existing research exhibits three critical methodological limitations that constrain our understanding of its pedagogical effectiveness.

First, previous studies have predominantly employed single-outcome measures, focusing either on linguistic competence (El Shazly, 2021; Qiao & Zhao, 2023) or affective variables (Zou et al., 2023) in isolation. This fragmented approach overlooks the interdependent relationship between cognitive and psychological dimensions of language learning, particularly the role of resilience in sustaining speaking development over extended periods. No empirical study has systematically examined how AI integration simultaneously affects both speaking proficiency and learning resilience within the same cohort.

Second, existing research lacks specificity in speaking skill assessment. While studies report general improvements in "speaking ability" (Ahyarudin & Jamilah, 2024; Qiao & Zhao, 2023), they fail to disaggregate performance across critical speaking components—pronunciation, fluency, accuracy, and coherence. This methodological limitation prevents identification of which specific aspects of oral production benefit most from AI intervention, limiting instructional design implications.

Third, research has yet to systematically examine how AI integration affects both cognitive and affective dimensions of language learning simultaneously. Most studies focus on either linguistic outcomes or psychological factors, leaving unexplored the potential synergistic effects of AI on both speaking competence and learner resilience.

These methodological gaps collectively limit our understanding of AI's dual role in addressing both the technical and psychological challenges inherent in lower-intermediate speaking development. The present study addresses these limitations through a semester-long quasi-experimental design that systematically examines both speaking performance components and resilience dimensions, providing a more comprehensive and methodologically rigorous evaluation of AI-enhanced instruction.

Research questions

How does AI-enhanced blended teaching affect the speaking performance of lower-intermediate EFL students compared to traditional classroom instruction over an 18-week semester?

How does AI-enhanced blended teaching affect the learning resilience of lower-intermediate EFL students compared to traditional classroom instruction over an 18-week semester?

Methodology

Building on the identified research gap and the study's focus on AI assisted speaking instruction and learning resilience, this section outlines the methodological framework used to investigate the effects of AI enhanced blended learning. The following subsections detail the research design, participant selection, tool justification, and instructional procedures that shaped the intervention and data collection process.

Research design

This study adopted a quasi-experimental pretest–posttest design with both between-group (instructional method) and within-group (time: pretest → posttest) comparisons to examine the impact of AI-enhanced blended instruction on EFL learners' speaking proficiency and learning resilience. Two intact classes of lower intermediate students (A2-B1 level) from a private university in southern Taiwan participated in the study, with one class assigned to the experimental group and the other to the control group. Participants' English proficiency was determined by their institutional placement test scores (CSEPT 180–219) at the time of admission. The two classes shared comparable course structures and learning objectives, making them suitable for parallel instruction and post intervention comparison. The intervention lasted for an 18-week semester, allowing sufficient time for meaningful development and assessment of speaking performance and learning resilience.

The independent (between-group) variable in this quasi-experimental design was the instructional method, distinguishing the experimental group (AI-enhanced blended instruction) from the control group (conventional instruction). The dependent variables were speaking performance and foreign language learning resilience; each measured at two time points (pretest and posttest). These repeated measurements constituted the within-group variables, enabling examination of both between-group differences and within-group changes across the 18-week intervention.

Participants

Participants in this study were drawn from two intact English communication classes at a private university in southern Taiwan. A total of 86 students participated, with each class containing 43 students. Participants ranged in age from 18 to 20 years, with 31 males (36.0%) and 55 females (64.0%). Students represented diverse academic backgrounds

across the university's departments, including English, Translation and Interpreting, Japanese, French, German, Spanish, International Affairs, Foreign Language Instruction, Applied Chinese Language, and Southeast Asian Studies.

All participants were classified as lower-intermediate learners based on their institutional placement test scores (CSEPT 180–219), corresponding to the A2-B1 level. The College Student English Proficiency Test (CSEPT) Level 2 is a standardized assessment comprising listening and reading sections, with a total possible score of 360 points. Although CSEPT does not directly assess speaking ability, a pilot assessment conducted with a subset of students ($n=20$) confirmed that those scoring in the 180–219 range demonstrated lower-intermediate speaking proficiency, characterized by limited fluency, frequent hesitation, and difficulty maintaining coherent discourse. All participants completed the CSEPT placement test within six months prior to the study for course placement purposes.

These two intact classes were selected based on their alignment with the institutional teaching schedule and curriculum structure. Using intact classes allowed for seamless integration of the intervention into the existing academic program without disrupting regular instruction or breaching institutional protocols. One class was assigned to the experimental group (AI-enhanced blended instruction) and the other to the control group (conventional instruction) through random assignment at the class level.

To maintain fairness and methodological rigor, both classes were matched in terms of course content, instructional hours, and assessment practices. Although random assignment at the individual level was not feasible due to ethical and logistical constraints of school-based research, the comparable nature of the two groups helped reduce potential bias and supported the internal validity of the study. The sample size and demographic characteristics are representative of typical EFL undergraduate cohorts in similar tertiary contexts, enhancing the transferability of the findings.

AI tools and rationale for selection

The AI tools used in the experimental group were ChatGPT and Google Gemini. These tools were selected based on both pedagogical and practical considerations. First, both tools are freely accessible and already familiar to many university students, minimizing the need for additional technical training and reducing potential affective barriers. Second, each tool was selected to match the demands of specific task types. ChatGPT was used for all real time interactive speaking tasks, including chatbot simulations and oral scenario practices, due to its robust dialogue generation capabilities. In contrast, Gemini was employed to support pre speaking activities such as brainstorming and idea organization, which required content generation rather than real time interaction.

This-task based integration allowed learners to experience complementary functionalities across tools while supporting targeted learning outcomes. Although other AI applications such as speech to text platforms or pronunciation scoring systems were considered, they were excluded due to limited conversational adaptability, higher operational complexity, or additional subscription costs. By selecting accessible and functionally suitable tools, the study ensured a balance between instructional depth and learner feasibility.

Instructional procedures

To provide a clear and replicable framework, the instructional process for both groups was organized into structured teaching phases within each 100-min session, held twice weekly over an 18-week semester. The framework was designed to support continuous speaking development while ensuring comparability between AI-enhanced and conventional instruction methods. The complete instructional framework with time allocations and weekly progression is detailed in Table 1.

Instruments

Foreign language learning resilience scale (FLLRS)

This study adopts the 19-item Foreign Language Learning Resilience Scale (FLLRS) developed and validated by Guo & Li (2022). Originally consisting of 24 items, the scale was refined to 19 through exploratory and confirmatory factor analyses (EFA and CFA). The final instrument comprises three dimensions: Ego Resilience (ER), Metacognitive Resilience (MR), and Social Resilience (SR), which respectively reflect learners' emotional adaptability, strategic learning behaviors, and ability to seek and provide social support in language learning contexts.

Participants rate each item on a 7-point Likert scale (1 = Strongly Disagree to 7 = Strongly Agree). The FLLRS was selected for its strong theoretical alignment with the cognitive, emotional, and social dimensions of resilience relevant to EFL learners. Its factor structure and item wording are particularly suitable for tertiary-level learners in East Asian contexts. In the present study, the scale also demonstrated high internal consistency (Cronbach's $\alpha = .85$ at pretest; $\alpha = .88$ at posttest), confirming its reliability with this sample.

Speaking performance assessment

Speaking performance was assessed using a structured task administered before and after the intervention, consisting of picture description, opinion expression, and role-play activities. Each response was rated on four analytic dimensions—fluency, accuracy, pronunciation, and coherence—using a standardized 100-point rubric (25 points per dimension). Two trained raters, blind to group assignment, independently evaluated the recordings following a calibration session. Inter-rater reliability, calculated via the intraclass correlation coefficient (ICC), reached .85, indicating excellent agreement. The full rubric is provided in Appendix 2.

Data collection procedure

1. *Pretest*: Before the intervention, all participants ($n=86$) completed the FLLRS and the speaking performance assessment to establish baseline measures. An independent samples t-test revealed no significant differences between the experimental group ($n=43$) and control group ($n=43$) in both FLLRS scores (EG: $M=3.24$, $SD=0.52$; CG: $M=3.31$, $SD=0.48$; $t(84)=-.842$, $p=.403$) and speaking performance

Table 1 Integrated instructional framework with time allocation and weekly progression

Session phase	Time	Experimental group (AI-enhanced)	Control group (conventional)	Weekly progression
Topic introduction	0–15 min	Teacher introduces theme; students explore background using ChatGPT chat mode with guided prompts	Teacher-led discussion with multimedia; peer sharing of prior knowledge	Weeks 1–6: Simple descriptive topics (family, hobbies) Weeks 7–12: Comparative topics (cultures, opinions) Weeks 13–18: Complex problem-solving topics
Vocabulary & pronunciation	15–25 min	ChatGPT voice mode for pronunciation practice: input → listen → record → compare → repeat. Document 5 challenging words	Teacher-led pronunciation drills; choral repetition and peer practice	Weeks 1–6: Basic vocabulary with teacher templates Weeks 7–12: Extended vocabulary with guided prompts Weeks 13–18: Advanced terminology with independent usage
Idea development	25–40 min	Gemini generates structured talking points: "Create 5 main points about [topic] for intermediate learners." Students select and adapt 3 points	Small group brainstorming with graphic organizers; teacher vocabulary support	Weeks 1–6: Teacher-provided AI templates Weeks 7–12: Student-created prompts with guidance Weeks 13–18: Independent AI consultation
Speaking practice	40–70 min	Phase A (15 min): ChatGPT chat mode for preparation Phase B (15 min): Role-plays using ChatGPT voice mode; peer observation and rotation every 5 min	Phase A (15 min): Teacher-guided preparation with worksheets Phase B (15 min): Traditional pair/group activities with teacher monitoring	Weeks 1–6: Focus on pronunciation accuracy and basic fluency Weeks 7–12: Emphasis on accuracy and coherence Weeks 13–18: Integration of all speaking sub-skills
Reflection & revision	70–85 min	ChatGPT analyzes performance: "Help me improve these sentences [student input]. Focus on grammar and natural expression." Practice revised versions	Peer feedback with checklists; teacher-facilitated error analysis and correction	Weeks 1–6: Guided reflection with structured prompts Weeks 7–12: Semi-independent analysis with teacher support Weeks 13–18: Strategic self-assessment and planning
Consolidation	85–100 min	Digital learning logs via Gemini; submit AI interaction evidence (screenshots/audio)	Written reflection journals; teacher-led summary and goal setting	All weeks: Progressive autonomy in reflection and goal-setting skills

(EG: $M=72.45$, $SD=8.63$; CG: $M=71.89$, $SD=8.95$; $t(84)=0.567$, $p=.573$), indicating that both groups were comparable at the outset of the study.

2. *Intervention*: Over the 18-week semester, the experimental group participates in the AI-enhanced blended learning environment, while the control group receives traditional classroom instruction.
3. *Posttest*: After the intervention, participants in both groups complete the FLLRS and the speaking performance assessment again. The posttest scores are compared to pretest scores to assess changes in learning resilience and speaking performance.

Data analysis

1. *Reliability analysis*: For the speaking assessment, inter-rater reliability was calculated to ensure consistency in rating. For the FLLRS (Guo & Li, 2022), which measures foreign language learning resilience through 19 items across three dimensions (ego resilience, metacognitive resilience, and social resilience), previous research has demonstrated high reliability (Cronbach's $\alpha=.940$). In the present study, internal consistency of the FLLRS was also examined and confirmed, with Cronbach's alpha values of .85 at pretest and .88 at posttest, indicating good reliability for this sample.
2. *Descriptive statistics*: Descriptive statistics (mean, standard deviation) are calculated for pretest and posttest scores on the FLLRS and speaking performance assessment for both groups.
3. *Inferential statistics*:
 - o *ANCOVA (analysis of covariance)*: ANCOVA is used to examine differences in posttest scores between the experimental and control groups, controlling for pretest scores. In this analysis, the between-group factor was the instructional method (AI-enhanced vs. conventional), and the covariate was the corresponding pretest score. Within-group changes were further examined through paired-samples t-tests comparing each group's pretest and posttest means. This analysis assesses the effectiveness of AI-enhanced blended teaching on speaking performance and learning resilience, accounting for initial differences between participants.
 - o *Effect size*: Effect sizes are calculated to determine the magnitude of the intervention's impact on speaking performance and resilience. To control for Type I error inflation from multiple comparisons, a Bonferroni correction ($\alpha=.05/4=.0125$) was applied. The correction was applied to the subcomponent analyses of speaking performance and learning resilience.

Results

This study examined the effects of AI-enhanced blended teaching on speaking performance and learning resilience among lower-intermediate EFL students over an 18-week intervention period. Both between-group and within-group comparisons were conducted to provide a comprehensive understanding of the intervention's impact. Before

Table 2 Reliability coefficients for study instruments

Instrument	Reliability type	Coefficient	95% CI
Speaking assessment	Inter-rater (ICC)	.85	[.81, .88]
FLLRS (Pretest)	Internal (Cronbach's α)	.85	[.82, .88]
FLLRS (Posttest)	Internal (Cronbach's α)	.88	[.85, .91]

Table 3 Descriptive statistics for speaking performance scores by group and time

Group	Pretest ($M \pm SD$)	Posttest ($M \pm SD$)	Adjusted Posttest ($M \pm SE$)
Experimental	72.45 $\pm$ 8.63	76.84 $\pm$ 7.82	76.72 $\pm$.71
Control	71.89 $\pm$ 8.95	74.62 $\pm$ 8.34	74.54 $\pm$.73

interpreting the results, it was essential to ensure that all assumptions underlying the use of analysis of covariance (ANCOVA) were satisfied to maintain the validity of the statistical procedures.

First, the assumption of homogeneity of regression slopes was tested by including an interaction term between the treatment group and the pretest scores. The interaction was not statistically significant, $F(1, 83) = 1.45$, $p = .232$, indicating that the relationship between pretest and posttest scores was consistent across groups. Second, the normality of residuals was assessed using Shapiro–Wilk tests for each dependent variable, with all p -values exceeding .05, suggesting that the residuals were normally distributed. Third, Levene's test was used to examine the homogeneity of variances, and the results were non-significant for all outcome variables ($p > .05$), confirming that variances were equal across groups.

Taken together, these findings confirmed that the core assumptions of ANCOVA were adequately met, thereby justifying the use of this statistical method for evaluating post-intervention differences while controlling for pretest scores.

Reliability analysis

The reliability analysis confirmed that all assessment instruments possessed good psychometric properties, as shown in Table 2.

The Intraclass Correlation Coefficient (ICC) for the speaking assessment was .85, indicating excellent inter-rater reliability. The Cronbach's α for the FLLRS was .85 at pretest and .88 at posttest, demonstrating strong internal consistency for both time points. Having established the reliability of the instruments, the following sections present the effects of the intervention on speaking performance and learning resilience.

Effects on speaking performance

Overall speaking performance

ANCOVA revealed a significant overall effect of the AI-enhanced blended teaching intervention on speaking performance, $F(1, 83) = 6.76$, $p = .007$, partial $\eta^2 = .07$, indicating an effect size that slightly exceeds the conventional threshold for a medium

Table 4 ANCOVA results for speaking performance components

Component	<i>F</i> (1, 83)	<i>p</i> -value	Partial η^2	EG Adjusted <i>M</i> (<i>SE</i>)	CG Adjusted <i>M</i> (<i>SE</i>)
Pronunciation	6.84	.007	.07	77.13 (0.70)	75.45 (0.73)
Fluency	2.12	.135	.02	76.57 (0.72)	76.12 (0.74)
Accuracy	5.87	.010	.06	77.46 (0.71)	75.89 (0.75)
Coherence	4.23	.029	.05	76.82 (0.71)	75.63 (0.74)

Table 5 Descriptive statistics for learning resilience scores by group and time

Group	Pretest (<i>M</i> $\pm$ <i>SD</i>)	Posttest (<i>M</i> $\pm$ <i>SD</i>)	Adjusted Posttest (<i>M</i> $\pm$ <i>SE</i>)
Experimental	3.26 $\pm$ 0.53	3.72 $\pm$ 0.44	3.71 $\pm$ 0.05
Control	3.27 $\pm$ 0.51	3.54 $\pm$ 0.49	3.55 $\pm$ 0.05

effect, as defined by Cohen (1988). The descriptive statistics for both groups' pretest and posttest speaking scores, including adjusted means, are presented in Table 3.

Speaking performance components

To gain a more detailed understanding of the intervention's impact, ANCOVA analyses were conducted for each speaking performance component: pronunciation, fluency, accuracy, and coherence. The results are summarized in Table 4.

For pronunciation, ANCOVA revealed a significant group effect, $F(1, 83) = 6.84$, $p = .007$, partial $\eta^2 = .07$. The adjusted mean score for the experimental group (77.13, $SE = .70$) was higher than that of the control group (75.45, $SE = .73$), indicating a moderate effect size favoring the intervention. This suggests that the intervention may have facilitated more clearly articulated speech production among lower-intermediate learners.

In the area of accuracy, the experimental group again outperformed the control group. A significant group difference was found, $F(1, 83) = 5.87$, $p = .010$, partial $\eta^2 = .06$, with adjusted means of 77.46 ($SE = .71$) and 75.89 ($SE = .75$), respectively. The effect size approached the conventional threshold for a medium effect, indicating potential gains in learners' grammatical precision. Among the four components, only pronunciation and accuracy remained statistically significant after Bonferroni correction ($\alpha = .0125$).

For coherence, ANCOVA yielded $F(1, 83) = 4.23$, $p = .029$, partial $\eta^2 = .05$. Although the p -value for coherence (.029) was below the conventional .05 threshold, it did not meet the Bonferroni-adjusted level ($\alpha = .0125$), and thus should be interpreted with caution.

In contrast, no significant group difference was found in fluency, $F(1, 83) = 2.12$, $p = .135$, partial $\eta^2 = .02$. The experimental group had a slightly higher adjusted mean (76.57) compared to the control group (76.12), but the difference was not statistically meaningful. The implications of this nonsignificant result will be further explored in the discussion section.

Impact on learning resilience

Overall learning resilience

ANCOVA revealed a significant overall effect of the AI-enhanced blended teaching intervention on learning resilience, $F(1, 83)=6.76$, $p=.007$, partial $\eta^2=.07$. Descriptive statistics and adjusted posttest means are presented in Table 5.

Learning resilience dimensions

Table 6 presents the descriptive statistics and ANCOVA results for each dimension of the Foreign Language Learning Resilience Scale (FLLRS).

Ego resilience (ER)

The ANCOVA results indicated that the EG did not significantly differ from the CG in ego resilience posttest scores, $F(1, 83)=2.15$, $p=.093$, partial $\eta^2=.02$. After Bonferroni correction, this difference was not significant ($p>.0125$). Paired-samples t-tests revealed a significant improvement in the EG from pretest to posttest ($t(42)=3.20$, $p<.01$), whereas the CG did not show a significant change ($t(42)=1.85$, $p=.070$). This suggests that ego resilience may be relatively stable and less responsive to the educational interventions employed.

Metacognitive resilience (MR)

The EG demonstrated significantly higher gains in metacognitive resilience compared to the CG, $F(1, 83)=7.42$, $p=.005$, partial $\eta^2=.08$. This difference remained significant after Bonferroni correction ($p<.0125$). Within-group analyses showed that both EG ($t(42)=6.80$, $p<.001$) and CG ($t(42)=3.50$, $p<.001$) experienced significant improvements, with the EG showing a more pronounced enhancement of 0.64 points over the CG. These findings suggest that while traditional instruction can enhance metacognitive resilience, AI-enhanced teaching provides additional benefits.

Social resilience (SR)

The ANCOVA results revealed that the EG had significantly higher posttest scores in social resilience compared to the CG, $F(1, 83)=5.95$, $p=.010$, partial $\eta^2=.06$. This difference remained significant after Bonferroni correction ($p<.0125$). Paired-samples t-tests indicated significant within-group improvements for both EG ($t(42)=7.10$, $p<.001$) and CG ($t(42)=4.00$, $p<.001$), with the EG achieving a greater gain of .50 points over the

Table 6 Descriptive statistics and ANCOVA results for FLLRS dimensions

Dimension	Group	Pretest ($M \pm SD$)	Posttest ($M \pm SD$)	$F(1, 83)$	p -value	Partial η^2
Ego resilience (ER)	EG	3.24 $\pm$ 0.52	3.48 $\pm$ 0.48	2.15	.093	.02
	CG	3.31 $\pm$ 0.48	3.42 $\pm$ 0.51			
Metacognitive resilience (MR)	EG	3.18 $\pm$ 0.57	3.82 $\pm$ 0.43	7.42	.005	.08
	CG	3.22 $\pm$ 0.54	3.58 $\pm$ 0.49			
Social resilience (SR)	EG	3.35 $\pm$ 0.49	3.85 $\pm$ 0.41	5.95	.010	.06
	CG	3.28 $\pm$ 0.51	3.62 $\pm$ 0.46			
Overall	EG	3.26 $\pm$ 0.53	3.72 $\pm$ 0.44	6.76	.007	.07
	CG	3.27 $\pm$ 0.51	3.54 $\pm$ 0.49			

CG. These results indicate that while both teaching approaches effectively develop social resilience, the AI-enhanced approach offers moderate additional advantages.

Summary of findings

The results indicate that both traditional and AI-enhanced blended teaching approaches contribute to improvements in speaking performance and learning resilience among lower-intermediate EFL students. Specifically:

- *Speaking performance:* The AI-enhanced blended teaching intervention led to significant improvements in pronunciation and accuracy compared to traditional instruction, with effect sizes indicating small-to-medium practical significance. Fluency improved similarly in both groups, suggesting that traditional methods are as effective as AI-enhanced approaches for this component.
- *Learning resilience:* The AI-enhanced approach significantly enhanced metacognitive and social resilience, whereas ego resilience remained largely unaffected. The effect sizes for these improvements ranged from small to medium, highlighting meaningful enhancements in students' ability to manage and adapt to language learning challenges.

These findings suggest that AI technology can effectively complement traditional teaching methods in EFL instruction, particularly in areas such as pronunciation development and the cultivation of metacognitive and social resilience. The differential impacts across various measures underscore the potential for strategically targeting specific aspects of language learning and resilience development through AI-enhanced blended teaching.

Ethical considerations

All participants provided informed consent prior to participation, and the study received approval from the Institutional Review Board (IRB) of Wenzao Ursuline University of Languages. Data confidentiality and anonymity were strictly maintained throughout the research process.

Discussion

This study investigated the effects of AI assisted blended instruction on speaking proficiency and learning resilience among lower intermediate EFL learners. The findings revealed that AI integration produced uneven yet meaningful gains across specific sub-skills and resilience dimensions, highlighting the differentiated mechanisms through which such technologies influence language learning outcomes.

Speaking abilities

Pronunciation and accuracy gains: evidence of targeted improvement

The experimental group showed significant improvements in pronunciation and oral accuracy through immediate feedback from ChatGPT and Gemini during speaking tasks. Students practiced native-like speech patterns, identified mispronunciations, and rehearsed language in communicative contexts. This approach differs markedly from existing research: Nutprapha & Dennis (2024) documented pronunciation gains through Speechling's repetition exercises focusing on isolated sounds, while the present

study embedded pronunciation practice within conversational contexts where feedback emerged naturally during communication attempts. Similarly, Gharamah et al. (2024) examined phonetic instruction through teacher-led sessions providing explicit phonological rules, whereas this study integrated pronunciation work within task-based interactions where learners discovered patterns through meaningful exchanges.

Beyond pronunciation, the study captured improvements in grammatical accuracy during oral production, with real-time feedback addressing errors in tense marking, subject-verb agreement, and word order as they occurred in conversation. This dual focus contrasts with Prahaladaiah & Thomas (2024), who examined pronunciation separately from grammatical features, reflecting traditional compartmentalized approaches to oral skills development. Iskandar et al. (2024) implemented project-based learning with teacher scaffolding, but feedback timing depended on teacher availability and class schedules. Technology-mediated feedback in the present study occurred immediately following each utterance, creating distinctly different learning conditions. These findings suggest that pronunciation and grammatical accuracy develop synergistically when feedback addresses both dimensions during meaningful communication, challenging conventional separation of these skills in language instruction.

Limited impact on fluency and coherence: the paradox of real-time intervention

The study revealed differential effects on fluency and coherence, with neither dimension achieving statistically significant improvements after Bonferroni correction. Fluency showed no significant between-group differences ($p = .135$), contrasting markedly with pronunciation and accuracy gains. This finding aligns with Kormos's (2014) theoretical framework, which posits that fluency development requires automatization and sustained speech production without interruption. Real-time AI feedback, despite its pedagogical value, appears to create cognitive interruptions that impede natural speech flow. Zou et al. (2023) documented similar patterns, demonstrating that immediate corrective input increases cognitive processing demands and reduces oral fluidity. The present findings extend this understanding: while AI feedback facilitates accuracy and pronunciation development, it may simultaneously disrupt the temporal dynamics essential for fluent speech production.

Coherence demonstrated marginal improvement ($p = .029$) that failed to meet the adjusted significance threshold ($\alpha = .0125$), indicating limited AI impact on discourse organization. Current generative AI platforms effectively address sentence-level errors but appear less capable of supporting complex discourse structuring. Qiao & Zhao (2023) identified strategic planning and genre-specific instruction as key factors in coherence development—elements that spontaneous AI interactions may not sufficiently provide.

A critical consideration emerges from the interactional dynamics of AI systems. Current AI systems tend to provide standardized, unitary responses that lack the interpretive freshness and contextual sensitivity essential for effective language teaching (Morales Fernández, 2024). These platforms typically accept and expand upon learner contributions without challenging underlying assumptions or requesting justification for claims. In contrast, human interlocutors regularly question logical inconsistencies and demand evidence, creating cognitive challenges essential for deep learning. Unlike human instructors who can engage learners in dynamic, challenging dialogue that requires real-time

discourse organization, AI platforms typically generate predetermined response patterns that may not adequately push learners to develop sophisticated coherence skills.

This type of intellectually demanding interaction serves as a catalyst for sophisticated discourse development. The absence of such challenges in AI interactions may explain the limited coherence gains observed. While AI systems reduce communicative anxiety and encourage participation, they may inadvertently eliminate the intellectual demands that foster advanced discourse competence. The marginal coherence improvements likely reflect enhanced surface cohesion through transitional devices rather than fundamental advances in logical organization. These findings illuminate a pedagogical tension: AI support enhances linguistic accuracy while potentially limiting exposure to the cognitive challenges necessary for developing advanced oral proficiency.

Learning resilience

Overall enhancement through AI assisted instruction

The intervention resulted in statistically significant gains in overall learning resilience, with a moderate effect size. This finding echoes Guo & Li (2022), who emphasized AI's role in supporting learner persistence and affective engagement. Unlike earlier studies that treated resilience as a monolithic construct, this study adopted a multidimensional framework, identifying the specific areas most responsive to technological support. Learners in the experimental group benefited from goal tracking, formative feedback, and task personalization—features that promote self-regulation. These results suggest that AI's contribution extends beyond cognitive domains to affective and behavioral regulation. While studies such as El Shazly (2021) noted challenges in reducing anxiety, the present findings demonstrate that well-designed AI integration can support emotional adaptation and learner persistence.

Dimension specific patterns: metacognitive, social, and ego resilience

The AI-supported intervention selectively enhanced metacognitive and social resilience, while producing no significant gains in ego resilience. This dimension-specific outcome offers a refined view beyond general claims (e.g., Guo & Li, 2022) that AI broadly supports all resilience domains. Gains in metacognitive resilience likely stemmed from structured tasks that prompted learners to set goals, monitor progress, and adjust strategies—patterns consistent with Qiao & Zhao's (2023) findings on feedback-driven learning. Social resilience improvements may be attributed to AI-supported peer interaction that reduced anxiety and encouraged engagement, aligning with Vygotsky's (1978) sociocultural perspective that emphasizes learning through collaborative exchanges.

In contrast, ego resilience—linked to emotional adaptability—remained unchanged. This may reflect the limitations of AI tools in fostering internal emotional regulation, which often depends on sustained human rapport and affective modeling. Interestingly, the control group also demonstrated moderate gains in metacognitive resilience, suggesting that well-structured traditional instruction can similarly support strategic learning behaviors. These findings highlight that AI is most effective when used deliberately, targeting specific psychological traits rather than assuming uniform affective benefits. Taken together, they suggest that AI tools can enhance selected aspects of resilience, but meaningful development depends on purposeful integration and the complementary role of human instruction.

Conclusion

This study demonstrates that AI-assisted blended instruction can meaningfully enhance specific components of speaking performance, particularly pronunciation and accuracy, while also fostering metacognitive and social resilience among lower-intermediate EFL learners. These results extend theoretical perspectives from Social Cognitive Theory, the Technology Acceptance Model, and Sociocultural Theory, offering empirical evidence that AI tools can support both linguistic development and learner agency when used with clear instructional design. However, the selective nature of the observed outcomes, such as the limited gains in fluency and the stability of ego resilience, underscores that technological tools are not universally effective. Their impact depends on how purposefully they are integrated into the learning process.

For classroom practitioners, the findings provide actionable guidance on embedding AI tools into blended learning environments. Rather than functioning as replacements for teacher feedback, tools such as ChatGPT and Gemini are most effective when employed as strategic supports within structured task cycles. Instructors can, for example, use AI to facilitate pronunciation rehearsal prior to oral production, and then incorporate AI-generated feedback during the post-task reflection phase. To improve grammatical accuracy, AI can be used to assist learners in sentence reformulation or guided practice within dialogue-based tasks. Fostering learning resilience requires carefully designed sequences that prompt students to set personal learning goals, monitor their output, and revise their performance with the support of timely, targeted AI prompts. Moreover, teachers should model how to critically evaluate AI-generated suggestions and guide students in developing the ability to discern, adapt, and self-regulate, rather than relying passively on automated output. These strategies aim to cultivate long-term linguistic competence while preserving reflective and learner-centered instruction.

From a policy perspective, this study highlights the importance of institutional and governmental investment in the infrastructure and training necessary for responsible AI integration in language education. Curriculum design should be updated to include AI-mediated communicative tasks that promote both language proficiency and digital literacy. Teacher education programs need to include pedagogical frameworks for AI use, along with ethical considerations and classroom implementation strategies. To reduce inequality, policies must also address access disparities by ensuring that all students, regardless of socioeconomic status, have the necessary technological resources. Furthermore, localized innovation should be encouraged through school-based pilot programs that explore culturally and contextually relevant applications of AI.

Overall, these findings point to the need for a transition from generalized optimism about AI in education to a more deliberate, context-aware, and pedagogically grounded implementation. The educational value of AI does not lie in automation alone, but in its potential to amplify human instruction when embedded within meaningful tasks, ethical guidance, and inclusive systems. Future research should continue to investigate long-term outcomes, the differential impacts of various AI tools, and adaptive strategies that support equitable and sustainable integration in diverse learning environments.

Limitations and future research directions

Despite its contributions, this study has several limitations that merit discussion. First, the relatively small sample size ($n = 86$), drawn from two intact classes at a single university, may limit the generalizability of the findings. Although both groups were matched in terms of course structure and learner proficiency, and the use of intact classes upheld institutional ethics, the absence of randomization and the constrained sample scope reduce external validity. Future research should strive to include larger, randomized samples from multiple institutions and broader proficiency ranges to enhance representativeness.

Second, this study relied on self-reported data to assess learning resilience, which introduces the risk of response bias. While the instrument used was previously validated, self-perception measures may not fully capture learners' behavioral or emotional resilience under authentic communication pressure. Future studies should consider mixed-method designs incorporating classroom observations, learner logs, or interview data to triangulate self-reported outcomes.

In addition, the assessment of speaking performance was based on analytic human ratings rather than objective indices. Although this approach allowed for detailed qualitative judgment, it did not include quantitative measures of complexity, accuracy, or fluency (e.g., speech rate, error-free clause ratio, lexical diversity). Future research could incorporate such objective or automated indices to complement human evaluation and enhance the reliability of speaking assessment.

Third, the 18-week intervention, while substantial, focused on short-term learning gains. The extent to which these improvements persist over time remains unknown. Longitudinal follow-up studies with delayed posttests could determine whether AI-integrated learning environments support lasting changes in speaking proficiency and learner autonomy.

Furthermore, although significant improvement was observed in accuracy, coherence, and resilience, speaking fluency did not show statistically meaningful gains. This suggests that AI tools, while useful for scaffolding content and accuracy, may not provide sufficient real-time interactional flow or prosodic modeling required for fluency enhancement. Future interventions could incorporate AI tools designed specifically for prosody training or real-time peer-interactive tasks to better target fluency.

Lastly, this study focused exclusively on ChatGPT and Gemini, two accessible and widely adopted AI tools. While this ensured feasibility and relevance for the learner population, it limited the comparative scope of technological influence. Future research should examine a broader array of AI platforms, including adaptive speaking tutors, AI-based pronunciation coaches, and multimodal apps, to identify which tool features align best with specific linguistic outcomes.

By acknowledging these limitations and outlining concrete directions for future inquiry, this study contributes a transparent and critical perspective that supports further development of AI-enhanced pedagogical models in EFL contexts.

Appendix 1

See Table 7

Table 7 Items for the *FLLRS*

Factor	No	Items in English
ER	1	I am curious about the new knowledge when I study a foreign language
	2	I am regarded as a very energetic person when I study a foreign language
	3	I am happy to make a favorable impression on others when I use a foreign language
	4	Perseverance matters, as it would never be easy to master a foreign language for a short period of time
MR	5	I would keep trying different methods to improve my foreign language
	6	I would use the feedback to improve my foreign language
	7	I would change my plans for language learning
	8	I would learn from the excellent classmates to improve my foreign language
	9	I would try to think more about my strengths and weakness to help me study a foreign language better
	10	I would set clear goals for achievements when I study a foreign language
SR	11	I am determined when I study a foreign language
	12	When I am encountered with difficulties in foreign language learning, I would seek encouragement from my classmates
	13	When I am encountered with difficulties in foreign language learning, I would seek help from my teachers
	14	When my classmate encountered with difficulties in foreign language learning, I would give a timely hand to help him/her
	15	Through foreign language learning, I can establish close and secure relationship with classmates
	16	It would be more efficient to ask others for help when I am encountered with difficulties in foreign language learning
	17	It would be okay if there are people who do not like me, I still seek for help when I am encountered with difficulties in foreign language learning
	18	To well solve problems in foreign learning, I would try to make foreign friends if time permits
	19	I would try collaborate with my classmates to complete foreign language learning tasks

ER Ego resilience, *MR* metacognitive resilience, *SR* social resilience, (-) refers to the removal of item in the explanatory factor analysis

Appendix 2

See Table 8

Table 8 Speaking performance assessment rubric (total: 100 points)

Criteria	Excellent (21–25)	Good (16–20)	Fair (11–15)	Limited (6–10)	Poor (1–5)
Fluency	Speaks smoothly with natural pacing and no hesitation; minimal pauses	Generally fluent with occasional hesitation or repetition	Noticeable hesitation; uneven flow; some difficulty maintaining fluency	Frequent pauses, slow speech, or evident struggle to continue	Very limited fluency; frequent breakdowns and long pauses impede communication
Accuracy	Excellent control of grammar and word choice with few or no noticeable errors	Good control with occasional minor errors that do not interfere with meaning	Some noticeable grammatical or lexical errors that may slightly affect clarity	Frequent errors that interfere with meaning or require listener effort to understand	Persistent and serious errors that severely limit comprehensibility
Pronunciation	Clear, natural, and comprehensible pronunciation with accurate stress and intonation	Mostly clear and intelligible with minor pronunciation issues	Sometimes unclear; pronunciation or stress patterns occasionally cause misunderstanding	Frequent mispronunciations that hinder understanding	Very difficult to understand due to unclear or inaccurate pronunciation
Coherence	Ideas are logically organized; responses are well-structured with appropriate connectors	Mostly well-organized with minor issues in structure or transitions	Some difficulty organizing ideas clearly; transitions may be weak	Disorganized responses; frequent lapses in logic or coherence	No clear organization; ideas are disconnected or hard to follow

Rubric descriptors were adapted based on CEFR oral descriptors and validated EFL speaking rubrics

Authors contribution

L solely conducted the conceptualization, data collection, analysis, manuscript writing, and revisions. The author reviewed and approved the final version of the manuscript.

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Data availability

The data supporting the findings of this study are available upon request.

Declarations**Ethics approval and consent to participate**

This study was conducted in accordance with the ethical guidelines outlined by the British Educational Research Association (BERA) Ethical Guidelines for Educational Research (2018).

Competing interests

The authors declare no competing interests.

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References

- Ahyarudin, D. A., & Jamilah, J. (2024). Enhancing students' speaking skills: The impact of AI-based instructional tasks in the genre-based approach. *Formosa Journal of Sustainable Research*, 3(9), 2019–2032. <https://doi.org/10.55927/fjsr.v3i9.11506>
- Alahmari, A., & Alrabai, F. (2024). The predictive role of L2 learners' resilience in language classroom engagement. *Frontiers in Education*, 9, 1502420. <https://doi.org/10.3389/educ.2024.1502420>
- Albelihi, H. H. M. (2022). Intermediate English as a foreign language learners' formulaic language speaking proficiency: Where does the teaching of lexical chunks figure? *Frontiers in Psychology*, 13, 949675. <https://doi.org/10.3389/fpsyg.2022.949675>
- Bandura, A. (1986). Social foundations of thought and action: A social cognitive theory. *Prentice-Hall*.
- Chiu, T. K. (2021). Digital support for student engagement in blended learning based on self-determination theory. *Computers in Human Behavior*, 124, 106909. <https://doi.org/10.1016/j.chb.2021.106909>
- Cohen, J. (1988). Statistical power analysis for the behavioral sciences (2nd ed.). Lawrence Erlbaum Associates.
- Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, 13(3), 319–340.
- Ebadijalal, M., & Yousofi, N. (2023). The impact of mobile-assisted peer feedback on EFL learners' speaking performance and anxiety: Does language make a difference? *The Language Learning Journal*, 51(1), 112–130. <https://doi.org/10.1080/09571736.2021.1957990>
- El Shazly, R. (2021). Effects of artificial intelligence on english speaking anxiety and speaking performance: A case study. *Expert Systems*, 38(12667), 1–15. <https://doi.org/10.1111/exsy.12667>
- Gharamah, F. M. A., Putro, N. H. P. S., & Pohan, A. E. (2024). Improving english pronunciation through phonetics instruction in Yemeni EFL classrooms. *JOLLT Journal of Languages and Language Teaching*, 12(2), 930–940. <https://doi.org/10.33394/jollt.v12i2.10720>
- Guo, N., & Li, R. (2022). Measuring Chinese English-as-a-foreign-language learners' resilience: Development and validation of the foreign language learning resilience scale. *Frontiers in Psychology*, 13, 1046340. <https://doi.org/10.3389/fpsyg.2022.1046340>
- Hashemi, A., & Na, K. S. (2020). The effects of using blended learning in teaching and learning English: A review of literature. *The Eurasia Proceedings of Educational & Social Sciences*, 18, 173–179.
- Holstein, K., McLaren, B. M., & Aleven, V. (2018). Student learning benefits of a mixed-reality teacher awareness tool in AI-enhanced classrooms. In *Penstein Rosé et al. (Eds.), AIED 2018, LNAI 10947* (pp. 154–168). https://doi.org/10.1007/978-3-319-93843-1_12
- Iskandar, I., Dewanti, R., Sulistyaningrum, S. D., & Santosa, I. (2024). Scaffolding assignments to conciliate the disinclination to employ project-based learning of English pronunciation and autodidacticism. *International Journal of Language Education*, 8(2), 199–227. <https://doi.org/10.26858/ijole.v8i2.64087>
- Jouibar, B., Hadian, B., & Dalili, M. V. (2021). The effect of task difficulty on the quantity and quality of iranian lower-intermediate EFL learners' code-switching. *Journal of Language Horizons, Alzahra University*, 5(1), 145–166. <https://doi.org/10.22051/lghor.2020.32109.1331>
- Kim, T. Y., & Kim, Y. K. (2016). The impact of resilience on L2 learners' motivated behavior and proficiency in L2 learning. *Educational Studies*, 43(1), 1–15. <https://doi.org/10.1080/03055698.2016.1237866>
- Kim, T.-Y., & Kim, Y. (2021). Structural relationship between L2 learning motivation and resilience and their impact on motivated behavior and L2 proficiency. *Journal of Psycholinguistic Research*, 50, 417–436. <https://doi.org/10.1007/s10936-020-09721-8>
- Kormos, J. (2014). The cognitive processes of language learning. In C. Doughty & M. Long (Eds.), *The Handbook of Second Language Acquisition* (pp. 356–375). Wiley-Blackwell.

- Lai, Z.C.-C. (2025). The impact of AI-assisted blended learning on writing efficacy and resilience. *International Journal of Computer-Assisted Language Learning and Teaching*, 15(1), 1–22. <https://doi.org/10.4018/JCALLT.377174>
- Lantolf, J. P., & Thorne, S. L. (2006). *Sociocultural theory and the genesis of second language development*. Oxford University Press.
- Luthar, S. S., Cicchetti, D., & Becker, B. (2000). The construct of resilience: A critical evaluation and guidelines for future work. *Child Development*, 71(3), 543–562. <https://doi.org/10.1111/1467-8624.00164>
- Matazu, S. S. (2024). Impact of AI-blended learning and AI-personalized learning on undergraduate biology students' attitude and performance in climate change education. *Anchor University Journal of Science and Technology*, 5(1), 83–95. <https://doi.org/10.4314/aujst.v5i1.8>
- Morales Fernández, B. (2024). The linguistic effects of artificial intelligence on the adequacy, coherence, and cohesion of texts: What do we stop telling in the new digital age? In R. V. Benítez Rojas & F.-J. Martínez-Cano (Eds.), *Revolutionizing communication: The role of artificial intelligence* (1st ed., pp. 1–11). CRC Press. <https://doi.org/10.1201/9781003473633>
- Namaziandost, E., & Rezai, A. (2024). Artificial intelligence in open and distributed learning: Does it facilitate or hinder teaching and learning? *The International Review of Research in Open and Distributed Learning*. <https://doi.org/10.19173/irrodl.v25i3.8070>
- Nutrapha, K. D. (2024). Using AI-powered speech recognition technology to improve English pronunciation and speaking skills. *IAFOR Journal of Education: Technology in Education*, 12(2), 107–112.
- Ortega Ruipérez, B. (2022). The role of metacognitive strategies in blended learning: Study habits and reading comprehension. *RIED Revista iberoamericana de educación a distancia*. <https://doi.org/10.5944/ried.25.2.32056>
- Pan, M. (2024). An AI-enhanced teaching model using a novel deep learning approach. *Applied Mathematics and Nonlinear Sciences*, 9(1), 1–19. <https://doi.org/10.2478/amns-2024-2567>
- Prahaladaiah, D., & Thomas, K. A. (2024). Effect of phonological and phonetic interventions on proficiency in English pronunciation and oral reading. *Education Research International*, 2024, 9087087. <https://doi.org/10.1155/2024/9087087>
- Qiao, H., & Zhao, A. (2023). Artificial intelligence-based language learning: Illuminating the impact on speaking skills and self-regulation in Chinese EFL context. *Frontiers in Psychology*, 14(1255594), 1–15. <https://doi.org/10.3389/fpsyg.2023.1255594>
- Sanders, D. A., & Mukhari, S. S. (2024). Lecturers' perceptions of the influence of AI on a blended learning approach in a South African higher education institution. *Discover Education*, 3, 135. <https://doi.org/10.1007/s44217-024-00235-2>
- Schmidt, T., & Strasser, T. (2022). Artificial intelligence in foreign language learning and teaching: A CALL for intelligent practice. *Anglistik: International Journal of English Studies*, 33(1), 165–184. <https://doi.org/10.33675/ANGL/2022/1/14>
- Utami, A. R., & Amalia, L. L. (2024). Tertiary students' challenges toward the use of oral presentation in speaking class. *Basic and Applied Education Research Journal*, 5(1), 1–10. <https://doi.org/10.11594/baerj.05.01.01>
- Vygotsky, L. S. (1978). *Mind in society: The development of higher psychological processes*. Harvard University Press. (Originally published in Russian in 1934)
- Wang, L. (2021). The role of students' self-regulated learning, grit, and resilience in second language learning. *Frontiers in Psychology*, 12, 800488. <https://doi.org/10.3389/fpsyg.2021.800488>
- Wang, Y., Derakhshan, A., & Zhang, L. J. (2021). Researching and practicing positive psychology in second/foreign language learning and teaching: The past, current status and future directions. *Frontiers in Psychology*, 12, 731721. <https://doi.org/10.3389/fpsyg.2021.731721>
- Wang, Y., Derakhshan, A., & Rahimpour, H. (2022). Developing resilience among Chinese and Iranian EFL teachers: A multi-dimensional cross-cultural study. *Journal of Multilingual and Multicultural Development*. <https://doi.org/10.1080/01434632.2022.2042540>
- Zhang, B. (2022). The relationship between Chinese EFL learners' resilience and academic motivation. *Frontiers in Psychology*, 13, 871554. <https://doi.org/10.3389/fpsyg.2022.871554>
- Zou, B., Guan, X., Shao, Y., & Chen, P. (2023). Supporting speaking practice by social network-based interaction in artificial intelligence (AI)-assisted language learning. *Sustainability*, 15(4), 2872. <https://doi.org/10.3390/su15042872>

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